# Shibasis Roy

+1 (831) 440-6136 | Santa Monica, CA | shibasis.cmi@gmail.com | Linkedin | shibasisr.github.io

## SUMMARY

Physicist with 5+ years of research experience building machine learning classifiers, time-series models, and Bayesian statistical frameworks for high-dimensional physics data. Independently developed an end-to-end ML forecasting pipeline alongside research code adopted by CERN collaborators. 2+ years of university-level teaching experience.

## SKILLS & CERTIFICATIONS

**Technical Skills:** Python (NumPy, SciPy, Pandas, Scikit-learn, Statsmodels, PyTorch), Matplotlib, Seaborn, Mathematica, C++ (GEANT4 simulation toolkit), AI-assisted development (Claude Code), Git/GitHub, PyROOT, SQL
**Domains:** machine learning (decision trees, neural networks), time series analysis, statistical modeling, Bayesian inference, Monte Carlo simulation, multi-dimensional angular-distribution / dynamical modeling, symbolic computation & automated derivation, SU(3)-flavor symmetry
**Broader interests:** Applied Probability, Computational Physics & Scientific Computing, Workflow Automation, Technical Communication

## EDUCATION

**Ph.D. & M.Sc. in Physics**, Institute of Mathematical Sciences, Chennai, India (GPA: 8.9/10) **July 2015 – Sept 2021**
Thesis: SU(3)-flavor analysis of hadronic bottom baryon decays; Adviser: Prof. Rahul Sinha.
Junior Research Fellowship, Joint CSIR-UGC NET in Physical Sciences (2015).
Full scholarship awarded through JEST for basic science Integrated PhD program.

**B.Sc. (Honors) in Physics**, Chennai Mathematical Institute, Chennai, India (GPA: 8.40/10) **Aug 2010 – July 2013**
Full Scholarship, National Undergraduate Programme in Mathematical Sciences.
“INSPIRE” fellowship under Kishor Vaigyanik Protshahan Yojana (KVPY).

## EXPERIENCE

**LHCb Research Group**, Eotvos Lorand University, Budapest **Postdoctoral Researcher** **July 2025 – Present**
- Translated theoretical rare-decay models into Monte Carlo simulation code (GEANT4/C++, PyROOT), improving simulation efficiency and cutting computation time 10% through algorithmic optimization.
- Diagnosed and fixed an accuracy problem in a vertex-reconstruction algorithm for tau leptons, improving reconstruction efficiency 20% and cutting misreconstructed events 10%.
- Built and validated ML classifiers (decision trees, neural networks) for tau identification in multi-body decay channels, achieving 85% classification accuracy.
- Conducted data analysis, statistical modeling, and visualization using Python (NumPy, SciPy, Pandas), PyROOT, and GEANT4.
- Designed a quantitative framework to predict CP violation in baryons and shared the results with experimental collaborators at CERN.

**Handshake** Model Validation Expert (MOVE) **Physics AI Trainer** **June 2025 – July 2025**
- Evaluated AI models against rigorous physics reasoning, identifying majority-case (>50%) reasoning errors and correcting model approaches — judging whether an answer is actually correct, not just plausible.
- Partnered with cross-disciplinary experts to translate domain-specific correctness criteria into AI training signal.

**University of California, Santa Cruz** **Visiting Postdoctoral Researcher** **Oct 2023 – May 2025**
- Led two independent, data-driven research projects (Python, Mathematica) end-to-end: statistical analysis, cross-validation, hypothesis testing, and a custom Monte Carlo event-rate/detector-acceptance simulator.
- Delivered results at international conferences (APS DPF-Pheno, APS April Meeting, and others); findings adopted as reference predictions by experimental collaborators, garnering over 40 citations.

**Chennai Mathematical Institute**, Chennai **Postdoctoral Fellow** **April 2023 – March 2024**
- Built a data-driven parameter extraction framework in Mathematica, applying constrained optimization to fit theoretical models to experimental measurements.
- Attended a graduate-level financial mathematics course for data science students, gaining exposure to core quantitative methods.

**University of Calcutta**, Kolkata **Postdoctoral Research Associate** **Dec 2020 – Dec 2022**

- Led a data-intensive Bayesian analysis demonstrating that apparent theory-experiment discrepancies were statistically insignificant ($<2\sigma$), providing actionable guidance to experimental collaborators.

**Institute of Mathematical Sciences, Chennai** **Doctoral Research Fellow** **Aug 2015 – Dec 2020**

- Built a symbolic computation tool (Python, Mathematica) automating SU(3) group-theoretic decomposition of particle decay amplitudes, cutting a multi-day manual derivation process to minutes.
- Designed simulation frameworks to model multi-dimensional angular distributions, enabling robust parameter extraction from high-dimensional data.

## PROJECTS

**Aurora Visibility Prediction** | Python, Pandas, Scikit-learn, Statsmodels, XGBoost, InterpretML **June 2026 – Present**

- Independently developing an end-to-end ML pipeline to predict aurora visibility using 22,000+ citizen science observations (2014–2025) merged with NASA OMNI2 solar wind data (Kp index, Bz GSM, solar wind velocity) and open-source Open-Meteo weather data (cloud cover, moon phase).
- Engineered geomagnetic time-series features with leave-one-region-out validation, including 3-hourly activity buckets and daily disturbance indices (Ap); built automated ingestion and preprocessing pipelines.
- Training classification models to identify space-weather and sky conditions favorable for aurora sightings, targeting real-time prediction and notification applications.

## PUBLICATIONS


CP-violation and U-spin symmetry in four-body bottom baryon decays — S. Roy, *Phys.Lett.B 869, 139838* 2025
Joint explanation of the $B \to \pi K$ puzzle and $B \to K\nu\bar{\nu}$ excess — S. Roy et al., *Phys.Rev.D 111, 7, 075029* 2025
Measuring CP-violating phase in beauty baryon decays — S. Roy et al., *Phys.Rev.Lett. 128, 8, 081803* 2022
Complete analysis of all $B \to \pi K$ decays — S. Roy et al., *Phys.Rev.D 104, 9, 095025* 2021
Beauty baryon nonleptonic decays into decuplet baryons… — S. Roy et al., *Phys.Rev.D 102, 5, 053007* 2020
Nonleptonic beauty baryon decays and CP-asymmetries… — S. Roy et al., *Phys.Rev.D 101, 3, 036018* 2020
Radiative b-baryon decays to measure the photon…polarization — S. Roy et al., *Eur.Phys.J.C 79, 7, 634* 2019
Lepton mass effects and angular observables in $\Lambda_b^0 \to \Lambda^0(\to p\pi)\ell^+\ell^-$ — S. Roy et al., *Phys.Rev.D 96, 11, 116005* 2017